Ryan Zhenqi Zhou, Ph.D. Candidate

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WORK EXPERIENCE

Research Assistant, GeoAI Lab, University at Buffalo – SUNY, Buffalo 09/2021 – Present

Climate Disaster Impact Modeling

- Developed data-driven frameworks for assessing spatial and temporal impacts of climate disasters using human mobility data, remote sensing images, and 311 service request data
- [AI] Developed **Geographical Random Forests (GRF) in Python** which takes spatial index into consideration [paper]
- [AI] Applied **time series models (LSTM, Transformer), statistical models (SLM, SEM, GWR, MGWR), and machine learning models (SVM, RF, GRF) with explainable AI (SHAP)** to predict disaster risks and evaluate differential physical and social vulnerabilities of communities during climatic events [GitHub repository]
- Findings supported local agencies in disaster response and first-author paper and paper published and presented at the 2023 and 2025 AAG Conferences

Obesity Prevalence Estimation

- Processed 3GB+ human mobility data and derived diet and physical activity features using **Python**
- [AI] Implemented **models (OLS, GWR, RF, DNN, GRF)** to predict obesity prevalence with accuracy over 90% through cross validation [GitHub repository]
- First-author paper published and presented at the 2022 GEOMED Conference

Geo-Knowledge-Guided GPT Model

- [AI] Fused geo-knowledge with **GPT models (GPT-2 to GPT-4)** via prompt engineering and question-answering to extract and classify complex location descriptions
- Achieved over 40% improvement in accuracy compared to NER baselines [paper]

Geospatial Data Scientist | Consultant, World Bank - IFC, Washington, D.C. 05/2024 - Present

Healthcare, Climate, and Tourism Analysis and Modeling

- Processed, analyzed, and mapped large-scale datasets, including healthcare facility data, patient records, climate disaster data (floods, extreme heat, and droughts), tourism data, wealth index data, population data, and remote sensing images in low- and middle-income countries (LMICs) using **Python, ArcGIS Pro, and Map API**
- Conducted geospatial analysis of 1,000+ healthcare facilities in Kenya and Tanzania, linking quality scores with climate and economic data to generate outcome indicators for resilience and equity
- [AI] Built predictive models: **Random Forest and Linear Regression** for wealth and population estimation (95%+ accuracy). **XGBoost and Huff Model** for forecasting healthcare facility transactions and market demand. Developed **GPT**-based address-matching pipeline to standardize unstructured data
- Results informed resilience-focused reforms and were adopted by local providers to improve healthcare service delivery

Research Assistant, Weill Cornell Medical College, Cornell University, New York City 08/2024 - 06/2025

Real-World Evaluation of Cardiovascular Disease Risk Factors

- Collected and processed large datasets on cardiovascular disease, socioeconomic status, and demographic at the neighborhood level in New York City
- Applied fuzzy string matching to geocode patient data and aggregate it to neighborhood level for analysis and modeling
- Built **Python** pipelines for scalable and reproducible data processing workflows [GitHub repository]

TECHNICAL SKILLS

- **Programming/Tools:** Python, R, SQL, Git, ArcGIS Pro
- **Libraries:** NumPy, Pandas, GeoPandas, Rasterio, SciPy, Scikit-Learn, Keras, TensorFlow, Matplotlib, Seaborn, Bokeh
- **AI:** Regression (OLS, GWR, MGWR, etc.), Tree-based Models (RF, XGBoost, GRF), Neural Networks (DNN, etc.), Time Series Models (LSTM, Transformer, etc.), Explainable AI (SHAP), Generative AI (GPT), Clustering (K-means)
- **Applications:** AI, Healthcare, Climate Risk, Human Mobility, Big Data Analytics, GIScience, Remote Sensing

EDUCATION

University at Buffalo – SUNY, USA 9/2021 - 09/2026 (expected)

Ph.D. in GeoAI and Geospatial Data Science (STEM)

- **Data Science Coursework:** Machine Learning, Deep Learning, Time Series Modeling, Database, Data Visualization
- **Geographic Information System Coursework:** GeoAI, Geospatial Big Data Analytics, Remote Sensing, Cartography